

Answer Key & Teacher Notes

Session 1 – Volume of Rectangular Prisms • Session 2 – Missing Fractional Edges

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 • TEACHER NOTES

Volume of Rectangular Prisms

TALK PROMPTS

- How is counting layers of cubes the same as multiplying length $\times$ width $\times$ height?
- How many base layers stack up inside the locker? How do you know?
- Why is volume written in cubic units and not square units?

WATCH FOR

- Adding the three edges instead of multiplying them.
- Multiplying only two edges (finding base area) and calling it the volume.
- Writing square units (cm^2) for a volume.

PARTNER TASK

Two Cartons. Each partner finds one carton's volume ($18 \times 5 \times 11 = 990 \text{ cm}^3$ and $15 \times 11 \times 5 = 825 \text{ cm}^3$). Then explain which holds more and why the shared edges 11 and 5 make the comparison quick.

SESSION 2 • TEACHER NOTES

Missing Fractional Edges

TALK PROMPTS

- Why does dividing by the base area give back the missing edge?
- What happens to the missing edge if the volume stays the same but the base gets bigger?

WATCH FOR

- Dividing the volume by only one known edge.
- Treating $1\frac{1}{2}$ as 1 (or as 1.2) when multiplying.
- Writing a missing edge in cubic units (cm^3) instead of length units.

PARTNER TASK

Two Unknowns. A prism has a volume of 270 cm^3 . Partner A finds the height when it is 9 cm by 5 cm (6 cm); Partner B finds the width when it is 6 cm long and 5 cm tall (9 cm). Check each other by multiplying all three edges.

ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 • Form A	6.G.A.2	1	1	1	1	1	2	2	3	3	1	1
S1 • Form B	6.G.A.2	1	1	1	1	1	2	2	3	3	1	1
S2 • Form A	6.G.A.2	1	2	1	1	1	2	2	3	3	1	1
S2 • Form B	6.G.A.2	1	2	1	1	1	2	2	3	3	1	1

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

GUIDED NOTES

volume: The amount of **space** inside a solid.

unit cube: A cube with every edge **1** unit long.

base area: The area of the **bottom** face of a prism.

cubic units: Units used to measure **volume**.

- One layer of cubes holds length $\times$ **width** cubes.
- The height tells how many **layers** stack up.
- $V = l \times w \times h$, or $V = B \times h$.

Example: One layer: $6 \times 3 = 18$ cubes. The height is 4, so there are **4** layers. Multiply: $18 \times 4 = 72$. Answer: $V = 72 \text{ cm}^3$

YOU TRY

1. $V = 5 \times 2 \times 3 = 30 \text{ in}^3$
2. a. 60 m^3 b. $25 \times 4 = 100 \text{ ft}^3$

EXIT TICKET

1. a. 42 cm^3 b. 150 in^3
2. $V = 4 \times 2 \times 3 = 24 \text{ ft}^3$
3. **C** (120 m^3)

PRACTICE FORM A

1. a. $8 \times 6 \times 9 = 432 \text{ cm}^3$ b. $7 \times 4 \times 3 = 84 \text{ ft}^3$ c. $9 \times 6 \times 2 = 108 \text{ ft}^3$ d. $42 \times 7 = 294 \text{ in}^3$
2. a. $5 \times 4 = 20$ cubes b. 3 layers c. $20 \times 3 = 60 \text{ ft}^3$
3. a. $B = 16 \times 11 = 176 \text{ in}^2$ b. $V = 176 \times 5 = 880 \text{ in}^3$
4. $B = 12 \times 15 = 180 \text{ in}^2$; $V = 180 \times 48 = 8,640 \text{ in}^3$
5. $V = 26 \times 11 \times 2 = 572 \text{ ft}^3$
6. No. Ian multiplied only two edges (that is the base area). $V = 17 \times 13 \times 5 = 1,105 \text{ cm}^3$.
7. A: 990 cm^3 ; B: 825 cm^3 . Carton A holds 165 cm^3 more.
8. a. $60 \times 30 \times 40 = 72,000 \text{ cm}^3$ b. $72,000 \div 1,000 = 72 \text{ L}$ c. $60 \times 30 \times 35 = 63,000 \text{ cm}^3 = 63 \text{ L}$
9. Tomás added the edges. Volume multiplies them: $V = 15 \times 11 \times 4 = 660 \text{ ft}^3$.
10. **C** (60 cm^3)
11. **C** (cubic units)

Self-check word CAPACITY

PRACTICE FORM B

1. a. $9 \times 5 \times 6 = 270 \text{ cm}^3$ b. $8 \times 3 \times 2 = 48 \text{ ft}^3$ c. $10 \times 7 \times 2 = 140 \text{ ft}^3$ d. $36 \times 8 = 288 \text{ in}^3$
2. a. $6 \times 3 = 18$ cubes b. 4 layers c. $18 \times 4 = 72 \text{ ft}^3$
3. a. $B = 18 \times 12 = 216 \text{ in}^2$ b. $V = 216 \times 6 = 1,296 \text{ in}^3$
4. $B = 14 \times 12 = 168 \text{ in}^2$; $V = 168 \times 50 = 8,400 \text{ in}^3$
5. $V = 24 \times 12 \times 2 = 576 \text{ ft}^3$
6. No. Rosa multiplied only two edges (that is the base area). $V = 19 \times 11 \times 4 = 836 \text{ cm}^3$.
7. A: 960 cm^3 ; B: 840 cm^3 . Carton A holds 120 cm^3 more.
8. a. $50 \times 30 \times 40 = 60,000 \text{ cm}^3$ b. $60,000 \div 1,000 = 60 \text{ L}$ c. $50 \times 30 \times 30 = 45,000 \text{ cm}^3 = 45 \text{ L}$
9. Malik added the edges. Volume multiplies them: $V = 12 \times 9 \times 5 = 540 \text{ ft}^3$.
10. **C** (60 m^3)
11. **A** (base area)

Self-check word CONTAINS

GUIDED NOTES

edge: A **segment** where two faces of a solid meet.

base area: The **product** of the length and width of the base.

inverse operations: Operations that **undo** each other.

- $V = l \times w \times h$ works with **fractions** too.
- To find a missing edge, **divide** the volume by the product of the other two edges.
- Check: the three edges should **multiply** to the volume.

Example: Known edges: $6 \times 4 = 24 \text{ cm}^2$. Divide: $72 \div 24 = 3$. Check: $6 \times 4 \times 3 = 72$. Answer: height = **3 cm**

YOU TRY

1. $V = 8 \times 1/2 \times 3 = 12 \text{ in}^3$
2. a. $60 \div 12 = 5 \text{ m}$ b. 9 ft^3

EXIT TICKET

1. a. 15 cm^3 b. $84 \div 12 = 7 \text{ in}$
2. $10 \times 6 = 60$; $180 \div 60 = 3 \text{ cm}$
3. **A** (6 ft^3)

PRACTICE FORM A

1. a. $3/4 \text{ ft}^3$ b. 77 in^3 c. $336 \div 48 = 7 \text{ in}$ d. $96 \div 12 = 8 \text{ in}$
2. a. $9 \times 10 = 90 \text{ in}^2$ b. $990 \div 90 = 11 \text{ in}$ c. Yes; 11 in is less than 13 in.
3. $B = 13 \times 10 = 130 \text{ cm}^2$; $h = 1,170 \div 130 = 9 \text{ cm}$
4. $20 \times 16 \times 3/4 = 320 \times 3/4 = 240 \text{ in}^3$
5. $20 \times 12 \times 1 1/2 = 240 \times 1 1/2 = 360 \text{ in}^3$
6. $V = 22 \times 9 \times 1/2 = 99 \text{ ft}^3$; cost $99 \times 5 = \$495$
7. $30 \times 18 = 540$; $5,400 \div 540 = 10 \text{ in}$
8. a. $3/4 \times 1/2 \times 2 = 3/4 \text{ m}^3$ b. $3/4 \times 18 = \$13.50$ c. $1 1/2 \text{ m}^3$; doubling the height doubles the volume.
9. Hana divided by only one edge. Divide by the product of both known edges: $11 \times 9 = 99$, and $1,980 \div 99 = 20 \text{ cm}$. Check: $20 \times 11 \times 9 = 1,980$.
10. **A** (6 cm)
11. **A** (1 m^3)

Self-check word FRACTION

PRACTICE FORM B

1. a. $1 1/2 \text{ m}^3$ b. 54 in^3 c. $240 \div 30 = 8 \text{ in}$ d. $105 \div 15 = 7 \text{ in}$
2. a. $8 \times 9 = 72 \text{ in}^2$ b. $1,080 \div 72 = 15 \text{ in}$ c. No; 15 in is more than 13 in.
3. $B = 12 \times 7 = 84 \text{ cm}^2$; $h = 1,008 \div 84 = 12 \text{ cm}$
4. $24 \times 12 \times 3/4 = 288 \times 3/4 = 216 \text{ in}^3$
5. $18 \times 10 \times 1 1/2 = 180 \times 1 1/2 = 270 \text{ in}^3$
6. $V = 20 \times 8 \times 1/2 = 80 \text{ ft}^3$; cost $80 \times 6 = \$480$
7. $25 \times 20 = 500$; $4,500 \div 500 = 9 \text{ in}$
8. a. $3/4 \times 1/2 \times 4 = 1 1/2 \text{ m}^3$ b. $1 1/2 \times 20 = \$30$ c. 3 m^3 ; doubling the height doubles the volume.
9. Jonah divided by only one edge. Divide by the product of both known edges: $12 \times 10 = 120$, and $2,520 \div 120 = 21 \text{ cm}$. Check: $21 \times 12 \times 10 = 2,520$.
10. **A** (8 in)
11. **A** (2 ft^3)

Self-check word MULTIPLY

HOMEWORK PACKET • ANSWERS

same answers for the core packet and Level 1

SHEET 1 • VOLUME OF RECTANGULAR PRISMS

1. $4 \times 3 \times 2 = 24 \text{ cm}^3$
2. a. 80 in^3 b. 400 cm^3 c. 216 ft^3 d. 315 m^3 e. 180 m^3 f. 288 in^3
3. $V = 6 \times 4 \times 1 = 24 \text{ ft}^3$; $24 \div 2 = 12 \text{ bags}$
4. $B = 20 \times 10 = 200 \text{ in}^2$; $h = 3,000 \div 200 = 15 \text{ in}$
5. No. $A = 6 \times 4 \times 2 = 48 \text{ in}^3$ and $B = 3 \times 4 \times 4 = 48 \text{ in}^3$. They hold the same amount.

SHEET 2 • MISSING FRACTIONAL EDGES

1. $4 \times 3/2 \times 5/2 = 60/4 = 15 \text{ in}^3$
2. a. 24 ft^3 b. 28 m^3 c. $5/2 \times 12 = 30 \text{ cm}^3$ d. $9/4 \times 4 = 9 \text{ in}^3$ e. $4 \times 1/2 \times 5 = 10 \text{ ft}^3$ f. $6 \times 3/2 \times 3 = 27 \text{ cm}^3$
3. $3 \times 1 1/2 = 4 1/2 \text{ ft}^2$; $9 \div 4 1/2 = 2 \text{ ft}$
4. a. $4 \times 2 \times 3 = 24 \text{ cubes}$ b. $24 \times 1/8 = 3 \text{ in}^3$ (check: $2 \times 1 \times 1 1/2 = 3$)
5. **D** (15 in^3)
6. **A** (4 ft)